Table S5 Negative-class and pocket-correspondence controls

A. Negative-class replacement

Target pair	Direction	n (dual / single / neither)	AUROC single-target	AUROC neither	Difference (95% CI)
EGFR/HER2	Dual vs A-only	28 / 37 / 12	0.6274	0.7083	0.0809 [-0.1447, 0.2871]
EGFR/HER2	Dual vs B-only	28 / 32 / 12	0.2467	0.7470	0.5004 [0.3025, 0.6819]
JAK1/JAK2	Dual vs A-only	32 / 32 / 14	0.5908	0.7344	0.1436 [-0.0498, 0.3326]
JAK1/JAK2	Dual vs B-only	32 / 32 / 14	0.7485	0.7143	-0.0342 [-0.2238, 0.1410]
JAK1/TYK2	Dual vs A-only	32 / 32 / 14	0.6475	0.7321	0.0847 [-0.0979, 0.2578]
JAK1/TYK2	Dual vs B-only	32 / 32 / 14	0.4219	0.8058	0.3839 [0.1953, 0.5693]
PIK3CA/mTOR	Dual vs A-only	32 / 32 / 14	0.7363	0.7679	0.0315 [-0.0897, 0.1497]
PIK3CA/mTOR	Dual vs B-only	32 / 32 / 14	0.5801	0.6964	0.1164 [-0.0592, 0.2872]
AChE/BChE	Dual vs A-only	27 / 25 / 15	0.6341	0.6963	0.0622 [-0.1047, 0.2242]
AChE/BChE	Dual vs B-only	27 / 28 / 15	0.6442	0.5679	-0.0763 [-0.2019, 0.0469]
F2/F10	Dual vs A-only	32 / 32 / 14	0.4082	0.5045	0.0963 [-0.0466, 0.2538]
F2/F10	Dual vs B-only	32 / 32 / 14	0.3623	0.4888	0.1265 [-0.0172, 0.2755]
PPARG/PPARA	Dual vs A-only	32 / 32 / 14	0.6553	0.7567	0.1014 [-0.0724, 0.2727]
PPARG/PPARA	Dual vs B-only	32 / 32 / 14	0.6021	0.7054	0.1033 [-0.1187, 0.3121]
PPARA/PPARD	Dual vs A-only	32 / 32 / 14	0.6787	0.6518	-0.0269 [-0.2102, 0.1486]
PPARA/PPARD	Dual vs B-only	32 / 32 / 14	0.5107	0.4866	-0.0241 [-0.1915, 0.1482]

Panel A difference = neither minus single-target negative class. Panel B difference = corresponding minus noncorresponding pocket. All estimates use S1. Individual AUROC confidence intervals and ligand identifiers are retained in the complete source CSVs. Corresp., corresponding pocket; noncorresp., noncorresponding pocket. Grouped counts follow the class order in the column heading.

Table S5 Negative-class and pocket-correspondence controls

B. Corresponding and noncorresponding pockets

Target pair	Direction	n (dual / negative)	Corresp. AUROC	Noncorresp. AUROC	Difference (95% CI)	Additional exclusions
EGFR/HER2	Dual vs A-only	28 / 37	0.6274	0.6573	-0.0299 [-0.0936, 0.0271]	0
EGFR/HER2	Dual vs B-only	28 / 32	0.2467	0.2355	0.0112 [-0.0681, 0.0949]	0
JAK1/JAK2	Dual vs A-only	32 / 32	0.5908	0.5908	0 [-0.0771, 0.0703]	0
JAK1/JAK2	Dual vs B-only	32 / 32	0.7485	0.7012	0.0474 [-0.0278, 0.1338]	0
JAK1/TYK2	Dual vs A-only	32 / 32	0.6475	0.6250	0.0225 [-0.0898, 0.1397]	0
JAK1/TYK2	Dual vs B-only	32 / 32	0.4219	0.4634	-0.0415 [-0.1279, 0.0410]	0
PIK3CA/mTOR	Dual vs A-only	32 / 32	0.7363	0.6519	0.0845 [-0.0454, 0.2144]	0
PIK3CA/mTOR	Dual vs B-only	32 / 32	0.5801	0.5859	-0.0059 [-0.1230, 0.1094]	0
AChE/BChE	Dual vs A-only	27 / 25	0.6341	0.4681	0.1659 [0.0548, 0.2830]	0
AChE/BChE	Dual vs B-only	27 / 28	0.6442	0.5807	0.0635 [-0.0661, 0.1971]	0
F2/F10	Dual vs A-only	32 / 32	0.4082	0.3926	0.0156 [-0.0957, 0.1230]	0
F2/F10	Dual vs B-only	32 / 32	0.3623	0.3887	-0.0264 [-0.1465, 0.0967]	0
PPARG/PPARA	Dual vs A-only	32 / 32	0.6553	0.5928	0.0625 [-0.1035, 0.2275]	0
PPARG/PPARA	Dual vs B-only	32 / 32	0.6021	0.6909	-0.0889 [-0.2563, 0.0869]	0
PPARA/PPARD	Dual vs A-only	32 / 32	0.6787	0.4434	0.2354 [0.0693, 0.4043]	0
PPARA/PPARD	Dual vs B-only	32 / 32	0.5107	0.4561	0.0547 [-0.1436, 0.2441]	0

Panel A difference = neither minus single-target negative class. Panel B difference = corresponding minus noncorresponding pocket. All estimates use S1. Individual AUROC confidence intervals and ligand identifiers are retained in the complete source CSVs. Corresp., corresponding pocket; noncorresp., noncorresponding pocket. Grouped counts follow the class order in the column heading.